AGENTIC AI SYSTEMS ENGINEER · MULTI-AGENT RUNTIMES · POLICY-FIRST CONTROL PLANES

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Summary

Agentic AI systems engineer with 40+ years in IT and the last decade building AI. I architect and ship the layer that decides what an agent is allowed to do — capability contracts, policy gates, human approval, and audit trails that prove what happened — then build the runtime, the APIs, the data layer, and the operator interfaces around it. Currently doing that for a six-agent, HIPAA-regulated production pipeline. Co-Chair of the C2PA AI/ML Task Force, contributor to its Agentic Task Force, and O'Reilly author of four books.

1 · Agentic Systems

Agent capability abstraction layer. Architected the ports-and-adapters boundary between six production agents and every backend they depend on: 16 domain-named capabilities over 24 swappable implementations. Agents request "Clinical Treatment Guidelines," never a vendor or a model, so a backend swap costs zero agent code changes.

Typed contracts over silent failure. Designed a capability outcome envelope that returns explicit unavailability instead of throwing. A mis-provisioned backend became a loud configuration error instead of a quiet fallback, and agents can be fully wired and tested against capabilities that have no implementation yet.

Policy-first control plane. Least-privilege tool and data access, human approval gates on consequential actions, and "unable to assess" routed to a human rather than defaulting to a denial. Codified "no agent contains capability logic" as a project standard, then drift-audited every agent against it.

Explicit, recoverable execution. Moved all agent dispatch onto queue-based execution end to end, with durable workflow state, retry and repair paths, and trace sealing across the web-request-to-worker boundary.

Forensic observability. Hash-chained tamper-evident execution traces with chain verification, liveness and stall detection, live SSE streaming, per-step token accounting, and audit entries recording which capability ran and which model answered.

Agent provenance and injection defense. Created Free2PA, a public toolkit that verifies signed agent control files before they reach model context — allow, deny, and quarantine primitives for untrusted instructions. Contribute to C2PA's Agentic Task Force, applying C2PA to agentic systems with colleagues from Adobe, Microsoft, Google, OpenAI, Meta, Amazon, Sony, and the BBC.

Agent tooling for other engineers. 13 internal Claude Code skills including wiring code generators used as the anti-drift mechanism, plus PAM, a credential-free browser-driving QA agent that acceptance-tests user journeys over CDP and files field-level evidence.

Multi-Agent Orchestration · Custom Agent Harnesses & Runtimes · Model Context Protocol (MCP) · Tool Adapters & Function Calling · Structured Outputs · Durable Session & Workflow State · Context-Window Management · Human-in-the-Loop Approval · Interruption Handling & Recovery · Event-Driven Queue Dispatch · Prompt-Injection Defense · AgentOps / LLMOps

2 · AI & Machine Learning

Model trust as a contract. Integrated an Azure ML prediction model into the agent loop behind a capability, with a four-state canonical trust enum — validated, unvalidated, simulated, unavailable — read by the tool, the agent prompts, the operator panel, and the validation report, because a model with an AUC of 0.897 behind it and one with no validation evidence were producing identical-looking results.

Model provenance and drift. Built a model-provenance ledger with drift detection that catches a model changing underneath a running agent, and extended the audit chain so a prediction is attributable to the exact version that produced it.

Judgment about when not to use an LLM. Converted four deterministic agents to LLM-backed agents on Azure AI Foundry for extraction, summarization, and narrative — while every approval decision stayed deterministic. Guideline answers require corroborating citations; the model never renders the verdict.

Retrieval and governed context. RAG and model routing in production, portable agent memory, and clinical NLP extraction from unstructured referral text over FHIR, SNOMED, and ICD-10, with guideline provenance and temporal validity treated as a product contract.

Applied AI research. AI Researcher and ML Engineer for Friends of Justin since 2023, a non-profit focused on improving interactions between humans and AI models.

Retrieval-Augmented Generation (RAG) · Model Routing & Provider-Aware Execution · Model Evaluation & Validation Evidence · Model Provenance & Drift Detection · MLOps / Production ML Lifecycle · Clinical NLP & Entity Resolution · Ontology & Schema Design · Azure AI Foundry, Azure OpenAI, Azure ML · Deepgram Speech AI · Python (intermediate)

3 · Impact Beyond a Single Role

Sets standards other organizations adopt. Co-Chair of the C2PA AI/ML Task Force and contributor to its Agentic Task Force and to the Society of Motion Picture and Television Engineers (SMPTE) / Entertainment Technology Center (ETC) AI/ML Task Force. These groups publish normative and non-normative standards and guidance documents.

Teaches the field. Four O'Reilly books — *Natural Language and Search*, *Blockchain Tethered AI*, *AI and the Law*, *Blockchain as a Service* — plus the Radar article "AI's Opaque Box Is Actually a Supply Chain," which framed AI as a traceable supply chain years before it was a category.

Raises the engineers nearby. Mentored the student engineer behind RadioHead, an award-winning transcription agent now running at an NPR member station. Ran hackathons and training programs that turned non-engineers into shipping builders.

Has led organizations, not just projects. Ran a 20-person Lotus Notes and Java consulting firm; CTO of a web development company; principal developer inside enterprise product teams; founder of three ventures.

Wins in the open. Team lead for the winner of the University of Arkansas AI Innovation and Integration Challenge (2026). Led the team that won the IBM Watson Build Challenge North America (2017). Six-time IBM Champion.

Builds the full path alone. Agent runtime, APIs, PostgreSQL schema and migrations, telemetry, React/TypeScript operator interfaces, and CLI and headless surfaces — same systems, in production.

TypeScript, Node.js, React, Next.js · PostgreSQL, Prisma, Migrations · Multi-Tenant APIs, OpenAPI · Azure Service Bus, KEDA, SSE · AWS Agent Deployment & Migration · OpenTelemetry, App Insights · CI/CD, Layered Testing, Release Gates · Least-Privilege Access, RBAC, Managed Identity · Content Provenance (C2PA, SMPTE, ETC) · Technical Writing & Mentoring

Experience

Founder & Principal Engineer —Kilroy AI LLC · *2026 – Present*

- **Hidalga** (client engagement) — agent capability layer, policy-first control, forensic observability, and model trust contract for a six-agent HIPAA-regulated oncology prior authorization pipeline.
- **RadioHead** — autonomous broadcast transcription agent for an NPR member station; started on OpenClaw, migrated onto AWS. Built with a student collaborator; won a student competition.
- **Free2PA** (free2pa.org) — public provenance toolkit verifying signed agent control files before model context load.
- **Phyllis** (phyllis.bot) — multi-tenant fulfillment API for autonomous commerce agents; agents prepare orders, humans approve anything that spends money or ships goods.

Founder & AI Engineer —NYX NoCode · *2024 – Present*

- Platform where non-specialists describe an application in natural language and get a deployed React app — agent orchestration, model routing, RAG memory, code generation, and hosting, owned end to end for public school customers in production.

AI Researcher & ML Engineer —Friends of Justin · *2023 – Present*

- AI research for a non-profit dedicated to improving interactions between humans and AI models.

Co-Chair, AI/ML Task Force · Contributor, Agentic Task Force —Coalition for Content Provenance and Authenticity (C2PA) · SMPTE / ETC AI/ML Task Force ·*Current*

- Cross-industry standards work on AI provenance and content authenticity, applying C2PA — a standard for determining the edits and origin of content — to agentic systems.

CEO —File Baby ·*2024 – 2025*

- Founded and led an early C2PA Content Credentials application, turning an emerging content provenance standard into a shipping product.

Author —O'Reilly Media ·*2019 – 2024*

- Four books: *Natural Language and Search* (2024), *Blockchain Tethered AI* (2023), *AI and the Law* (2021), *Blockchain as a Service* (2019), plus the O'Reilly Radar article "AI's Opaque Box Is Actually a Supply Chain."

CEO & Technical Lead —Kilroy Blockchain ·*2016 – 2025*

- Led the team that built **RILEY**, winner of the IBM Watson Build Challenge North America (2017). Directed engineering for blockchain-backed workflow and case management systems where auditability was a core requirement.

Earlier Roles (Condensed)

- **CTO – Jamersan LLC** (2016) · **Magento Technical Lead – Amplifi Commerce** (2015–2016) · **Principal Applications Developer – CA Technologies** (2014–2015) · **Training & Documentation Consultant – Magento Inc.** (2010–2014), Magento U instructor and course author
- **Web Programmer, then Executive Director of United Cloud – Suarez Corporation Industries** (2009–2012) · **Webmaster – Kucinich for President** (2003–2004), the first US presidential election where the web was a factor
- **President – Data Now** (1991–1999), ran a 20-person Lotus Notes and Java consulting firm · **LAN Administrator – Bayer** (1990–1998), 200-user network · **Technology Coordinator – CIGNA** (1983–1991). IT career began in 1980 as a telex operator.

Recognition, Certifications & Education

Winner, University of Arkansas AI Innovation and Integration Challenge, team lead (2026) · IBM Watson Build Challenge Winner, North America (2017) · IBM Champion, six-time honoree (2020–2025)

AI Fluency for Students and Teaching the AI Fluency Framework — Anthropic (2025) · Venture Building — Builders + Backers (2025) · IBM Certifications: Watson Chatbot, RPA, Bluemix Essentials, Blockchain Essentials · FAA Private Pilot License

University of Arkansas, Sam M. Walton College of Business — Cloud Computing & Infrastructure, agent-focused (Spring 2026), 4.0 GPA; Quantum Computing (Fall 2026)

University of Arkansas — Studies in Music, 4.0 GPA (2023–Present) · **Hammel College** — Office Automation & Database Management (1981–1982), 4.0 GPA